

POCKETSMART AI

File 2 — Requirement Analysis

1. Purpose of the Requirement Analysis

This phase converts the PocketSmart AI concept into clear functional and technical requirements. The system must accept a user's budget and requirements, understand them using Generative AI, and return relevant recommendations in an understandable format.

2. Stakeholders and Users

Stakeholder / User	Requirement or Interaction
End User	Provides budget, category, preferences, and other requirements; receives recommendations.
System / AI	Interprets the request, applies constraints, and generates recommendation content.
Application Administrator	Maintains the application and supporting services where required.

3. Functional Requirements

ID	Requirement
FR-01	The system shall allow the user to enter a spending budget.
FR-02	The system shall allow the user to specify a product, service, or planning category.
FR-03	The system shall accept natural-language preferences and requirements.
FR-04	The system shall process the user's request using a Generative AI model.
FR-05	The system shall generate recommendations relevant to the user's stated budget and requirements.
FR-06	The system should explain recommendations in clear natural language.
FR-07	The system should support multiple use cases such as home decor, event planning, jewellery shopping, and general shopping.

4. Non-Functional Requirements

Category	Requirement
Usability	The interface should be simple enough for users to enter requirements without technical knowledge.
Performance	The application should return AI-generated results within a reasonable response time.
Scalability	The architecture should allow additional recommendation categories and services to be added later.
Reliability	The application should handle invalid or incomplete user input gracefully.
Security	API credentials and application configuration must not be exposed in the user interface or source repository.
Maintainability	AI, frontend, backend, and recommendation components should be organized so they can be updated independently.

5. Input Requirements

The minimum user input is expected to include a budget and the user's requirement or category. Additional preferences can be supplied in natural language to improve recommendation relevance.

Input	Example Purpose
Budget	Defines the maximum or intended spending amount.

Category	Identifies the area such as home decor, event planning, or jewellery.
Preferences	Describes style, quantity, quality, occasion, or other requirements.
Optional Constraints	Additional conditions that can narrow the recommendations.

6. Output Requirements

The output should present a manageable set of recommendations that are aligned with the user's budget and requirements. Each recommendation should be understandable to a non-technical user and, where applicable, include an explanation of its relevance.

7. Technical Requirements

Layer	Requirement / Technology Direction
AI	Generative AI using the Gemini API.
Application	A web application capable of accepting user input and displaying recommendations.
Data / Recommendation Layer	A mechanism for representing or retrieving product/service information for recommendation.
Security	API keys and credentials stored securely as environment configuration.

8. Requirement Summary

The requirement analysis establishes the minimum scope for PocketSmart AI: a web-based AI assistant that understands budget-constrained requests and generates relevant recommendations. The requirements can be used as the baseline for the next phase, Project Design.